

10.5.9

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Question

Draw a circle of radius 4 cm. Draw two tangents to the circle inclined at an angle of 60° to each other.

Solution

The parameters of the circle with center $\mathbf{0}$ are :

$$\mathbf{V} = \mathbf{I} \qquad \mathbf{u} = \mathbf{0} \qquad f = -16 \qquad (0.1)$$

Let the point from which tangent is being drawn be $\mathbf{p}$.

Let the point of contact be $\mathbf{r}$ and

$$\mathbf{r}^\top \mathbf{r} = 16 \qquad (0.2)$$

From the condition of tangency we get

$$\mathbf{r}^\top (\mathbf{r} - \mathbf{p}) = 0 \qquad (0.3)$$

$$\mathbf{p}^\top \mathbf{r} = \mathbf{r}^\top \mathbf{r} \qquad (0.4)$$

$$\mathbf{p}^\top \mathbf{r} = 16 \qquad (0.5)$$

If the angle between the tangents is 60° then the angle between the normals at the points of contact is 120° .

Solution

Therefore,

$$\cos\left(\frac{120^\circ}{2}\right) = \frac{\mathbf{p}^\top \mathbf{r}}{\|\mathbf{p}\| \|\mathbf{r}\|} \quad (0.6)$$

$$\|\mathbf{p}\| = 8 \quad (0.7)$$

$$\mathbf{p}^\top \mathbf{p} - 64 = 0 \quad (0.8)$$

Therefore the locus of point $\mathbf{p}$ is a circle with center $\mathbf{0}$ and radius 8 cm.

Consider point $\mathbf{P} = \begin{pmatrix} 8 \\ 0 \end{pmatrix}$ (lies on the locus) from which tangents are drawn.

Let the tangent equation passing through $\mathbf{P}$ be

$$\mathbf{x} = \mathbf{P} + k\mathbf{m} \quad (0.9)$$

Solution

Finding the point of contact :

$$g(\mathbf{x}) = \mathbf{x}^\top \mathbf{x} - 16 \quad (0.10)$$

$$(\mathbf{P} + k\mathbf{m})^\top (\mathbf{P} + k\mathbf{m}) - 16 = 0 \quad (0.11)$$

$$k^2 \mathbf{m}^\top \mathbf{m} + 2k \mathbf{P}^\top \mathbf{m} + \mathbf{P}^\top \mathbf{P} - 16 = 0 \quad (0.12)$$

$$k^2 \mathbf{m}^\top \mathbf{m} + 2k \mathbf{P}^\top \mathbf{m} + g(\mathbf{P}) = 0 \quad (0.13)$$

$$(0.14)$$

As the tangent intersects the conic at only one point(the point of contact),the discriminant for the quadratic in k is equal to 0

$$g(\mathbf{P}) = 48 \quad (0.15)$$

$$\mathbf{m}^\top \begin{pmatrix} -16 & 0 \\ 0 & 48 \end{pmatrix} \mathbf{m} = 0 \quad (0.16)$$

$$\mathbf{Q} = \begin{pmatrix} -16 & 0 \\ 0 & 48 \end{pmatrix} \quad (0.17)$$

$$(0.18)$$

Solution

As $\mathbf{Q}$ is an upper triangular matrix , the eigen values are the diagonal entries :

$$\lambda_1 = -16 \qquad \lambda_2 = 48 \qquad (0.19)$$

Applying eigen value decomposition for $\mathbf{Q}$

$$\mathbf{Q} = \mathbf{XDX}^T \qquad (0.20)$$

$$\mathbf{D} = \begin{pmatrix} -16 & 0 \\ 0 & 48 \end{pmatrix} \qquad (0.21)$$

$\mathbf{X}$ is an orthogonal matrix whose columns are the corresponding normalized eigenvectors of $\mathbf{Q}$

As $\mathbf{Q}$ is a diagonal matrix ,

$$\mathbf{X} = \mathbf{I} \qquad (0.22)$$

Solution

From (0.16) ,

$$\mathbf{m}^T \mathbf{X} \mathbf{D} \mathbf{X}^T \mathbf{m} = 0 \quad (0.23)$$

$$\mathbf{z} = \mathbf{X}^T \mathbf{m} \quad (0.24)$$

$$\mathbf{z}^T \mathbf{D} \mathbf{z} = 0 \quad (0.25)$$

$$(\lambda_1 \quad \lambda_2) \begin{pmatrix} -16 & 0 \\ 0 & 48 \end{pmatrix} \begin{pmatrix} \lambda_1 \\ \lambda_2 \end{pmatrix} = 0 \quad (0.26)$$

$$\frac{\lambda_1}{\lambda_2} = \pm \sqrt{3} \quad (0.27)$$

Solving for $\mathbf{m}$,

$$\mathbf{I} \mathbf{m} = \mathbf{z} \quad (0.28)$$

$$\mathbf{m} = \begin{pmatrix} \lambda_1 \\ \lambda_2 \end{pmatrix} \quad (0.29)$$

$$\mathbf{m} = \begin{pmatrix} 1 \\ \frac{\lambda_2}{\lambda_1} \end{pmatrix} \quad (0.30)$$

Solution

From (0.27) , the direction vectors for the tangents are given as :

$$\mathbf{m}_1 = \begin{pmatrix} 1 \\ \frac{1}{\sqrt{3}} \end{pmatrix} \quad \mathbf{m}_2 = \begin{pmatrix} 1 \\ -\frac{1}{\sqrt{3}} \end{pmatrix} \quad (0.31)$$

The normal vectors for the tangents are given as :

$$\mathbf{n}_1 = \begin{pmatrix} -\frac{1}{\sqrt{3}} \\ 1 \end{pmatrix} \quad \mathbf{n}_2 = \begin{pmatrix} \frac{1}{\sqrt{3}} \\ 1 \end{pmatrix} \quad (0.32)$$

The points of contacts are given as :

$$\mathbf{r}_i = \pm r \frac{\mathbf{n}_i}{\|\mathbf{n}_i\|} \quad (0.33)$$

From (0.5) , $\mathbf{P}^\top \mathbf{r} = 16$, so the points of contact are :

$$\mathbf{r}_1 = \begin{pmatrix} 2 \\ 2\sqrt{3} \end{pmatrix} \quad \mathbf{r}_2 = \begin{pmatrix} 2 \\ -2\sqrt{3} \end{pmatrix} \quad (0.34)$$

Plot

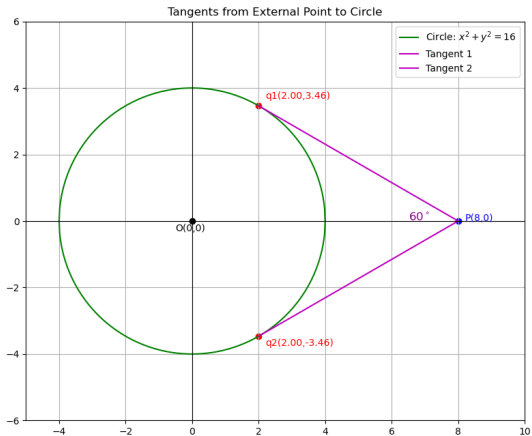


Fig : Circle and Tangents